American Computer Science League

2024 Finals ● Short Problems ● Intermediate/Classroom Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$A(\overline{B}C) + \overline{ABC} + (\overline{A} + \overline{B})C$$

A. 0
B. 1
C. 3
D. 5
E. None of the above

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{\overline{B}A} + (B \oplus C)$$

A. $\overline{A} + B + \overline{B}\,\overline{C}$
B. $\overline{A} + \overline{B} + BC$
C. $A + B + \overline{B}C$
D. $\overline{A} + B + \overline{B}C$
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

01100 OR (LSHIFT-1 (RCIRC-2 01011))
AND (LCIRC-2 (RSHIFT-1 01010))

A. 10100
B. 00111
C. 11100
D. 01010
E. None of the above

4. Bit-String Flicking

How many different values of X, a bit-string of 5 bits, make the following equation TRUE?

(RCIRC-2 01101) AND (LSHIFT-1 X) =
(LSHIFT-2 (NOT 10111 OR X))

A. 0
B. 2
C. 4
D. 8
E. None of the above

5. Recursive Functions

Find $f(18)$, given:

$$f(x) = \begin{cases} f(x-3)+1 & \text{if } x \geq 13 \\ 2*f(x+1)-2 & \text{if } 10 < x < 13 \\ 2*x-4 & \text{if } x \leq 10 \end{cases}$$

A. 16
B. 17
C. 32
D. 33
E. None of the above

6. Recursive Functions

Find $f(40, 28)$, given:

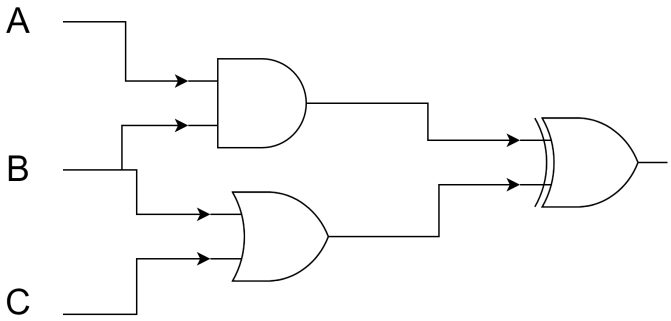
$$f(x,y) = \begin{cases} f\left(\left[\frac{x}{3}\right], \left[\frac{y}{2}\right]\right)+1 & \text{if } x \geq 15 \\ f(y+4,\ x-2) & \text{if } 8 < x < 15 \\ x+y & \text{if } x \leq 8 \end{cases}$$

Note: $[a]$ is the greatest integer less than or equal to a

A. 10
B. 11
C. 12
D. 13
E. None of the above

7. Digital Electronics

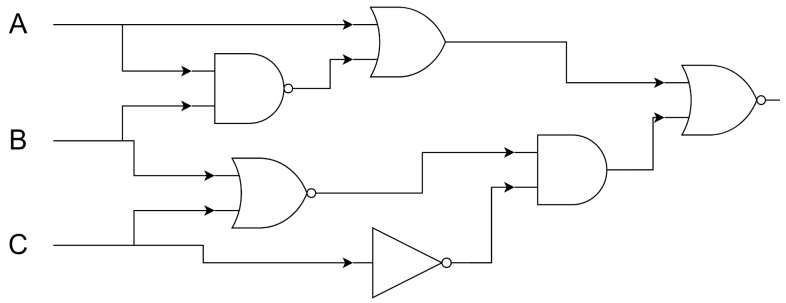
How many ordered triples make the following circuit TRUE?



A. 2
B. 4
C. 6
D. 8
E. None of the above

8. Digital Electronics

Simplify the Boolean expression that represents the following digital circuit:



A. 0
B. $\overline{B}$
C. $\overline{C}$
D. $\overline{B}\,\overline{C}$
E. None of the above

9. Prefix-Infix-Postfix

Define two new operators:
$ is the maximum of its operands
is the minimum of its operands

Evaluate the following prefix expression, if all numbers are single digits:

− * # 2 ^ 3 2 $ * 4 2 3 − ^ 6 2 * 3 4 0

A. 16
B. 0
C. -8
D. 8
E. None of the above

10. Prefix-Infix-Postfix

Convert the following infix expression to postfix.

$$\frac{5(2A+4)}{2B-3} - 7$$

A. 52A*4+*2B*3−/7−
B. 52A4*+*2B*3−/7−
C. 52A*4+*/2B*3−7−
D. 52A*4*+2B*3−/7−
E. None of the above

11. Computer Number Systems

What is the value of the following expression in hexadecimal?

$77777_8 + 7777_8 - 777_{16}$

A. 107776
B. 8FFE
C. 8887
D. 34951
E. None of the above

12. Computer Number Systems

What is the final octal digit of 3_8 ^ 7_{10}?

A. 0
B. 1
C. 3
D. 5
E. None of the above

13. Data Structures

Build a binary search tree for:

DISHWASHER

What is the preorder traversal of the resulting tree?

A. DAIHHESWSR
B. DAIHSHSWER
C. DIASWSRHHE
D. DAIHHESSRW
E. None of the above

14. Data Structures

Consider the following snippet of pseudo-code that processes a string:

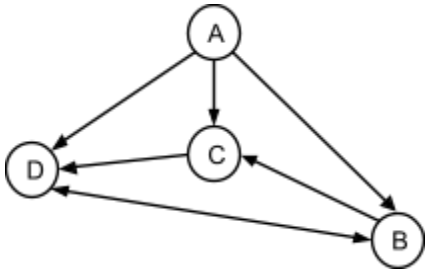
```
for each character ch in string
    if ch is "{" or "[" or "(" then
        push(ch)
    else if ch is "}" or "]" or ")" then
        other = pop()
```

Which string will result in a non-empty stack after the code is run?

A. {[(x+y)*(x-y)](x/y)}
B. {[(x+2)][(x-3)(4x)]}
C. {[x*y)^2+(x/y]^3}-2
D. {[(x-2y[(3x+4]](2x]}
E. None of the above

15. Graph Theory

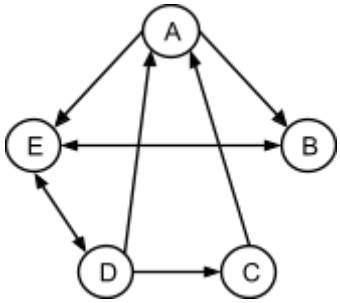
Given the directed graph below, which vertex has the most paths of length 4 starting from that vertex?



A. A
B. B
C. C
D. D
E. More than 1 of them

16. Graph Theory

How many cycles are there in the following directed graph?



A. 3
B. 4
C. 5
D. 6
E. None of the above

17. What Does This Program Do?

What is the output when the following program is executed?

```
s1 = 1
for x = 2 to 10
    s1 = s1 + x^2
    y = 1
    s2 = 1
    while s2 < s1
        y = y + 1
        s2 = s2 + y
    end while
    if s1 == s2 then
        output y
    end if
next x
```

A. 2, 11, 14
B. 1, 10, 13
C. 10, 13
D. 10
E. None of the above

18. LISP

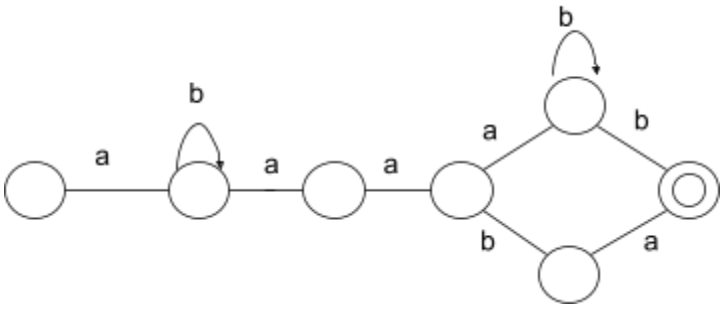
Evaluate the following LISP expression:

(REVERSE (CADADR '(1 (2 (3 2 (1))) (2 ((2) 1)))))

A. ((1) 2 3)
B. (1 2 3)
C. (1 (2))
D. (3 2 1)
E. None of the above

19. FSAs

Which of the following strings are accepted by the following FSA?



- a b b a a b
- a a a b a
- a b a a b a b
- a b b b a a a
- a b b a a b a a

A. all of them
B. exactly 1 of them
C. exactly 3 of them
D. all but 1 of them
E. None of the above

20. Assembly Language

What is printed when the following program is run?

```
NUM         DC          10101110
ANS         DC          0
POW         DC          1
SUM         DC          0
            LOAD        NUM
TOP         BE          STOP
            DIV         =10
            STORE       ANS
            MULT        =10
            SUB         NUM
            BL          CONT
            BU          BACK
CONT        LOAD        ANS
            MULT        =-10
            ADD         NUM
            MULT        POW
            ADD         SUM
            STORE       SUM
BACK        LOAD        POW
            MULT        =2
            STORE       POW
            LOAD        ANS
            STORE       NUM
            BU          TOP
STOP        PRINT       SUM
            END
```

A. 5
B. 87
C. 174
D. 255
E. None of the above